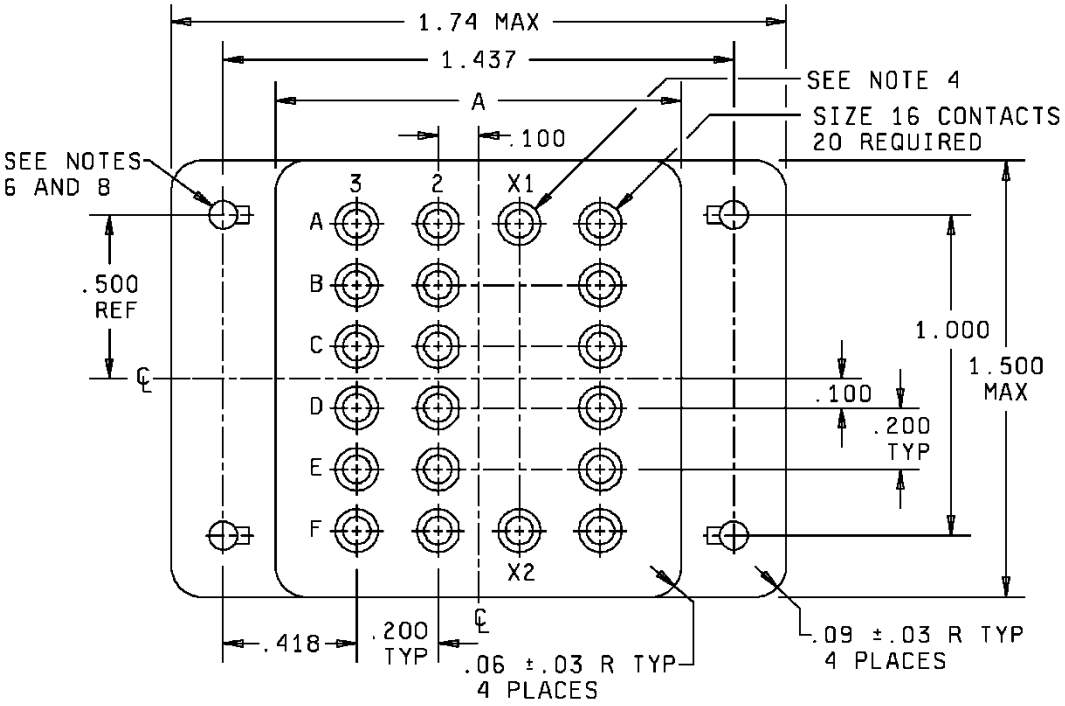
INCH-POUND
MIL-DTL-12883/47D
w/AMENDMENT 2
21 January 2020
SUPERSEDING
MIL-DTL-12883/47D
w/AMENDMENT 1
3 Aug 2015

DETAIL SPECIFICATION SHEET

SOCKET, PLUG-IN ELECTRONIC COMPONENTS, FOR RELAYS, 6-POLE, 10 AMPERES (MIL-PRF-6106 AND MIL-PRF-83536)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification and MIL-DTL-12883.



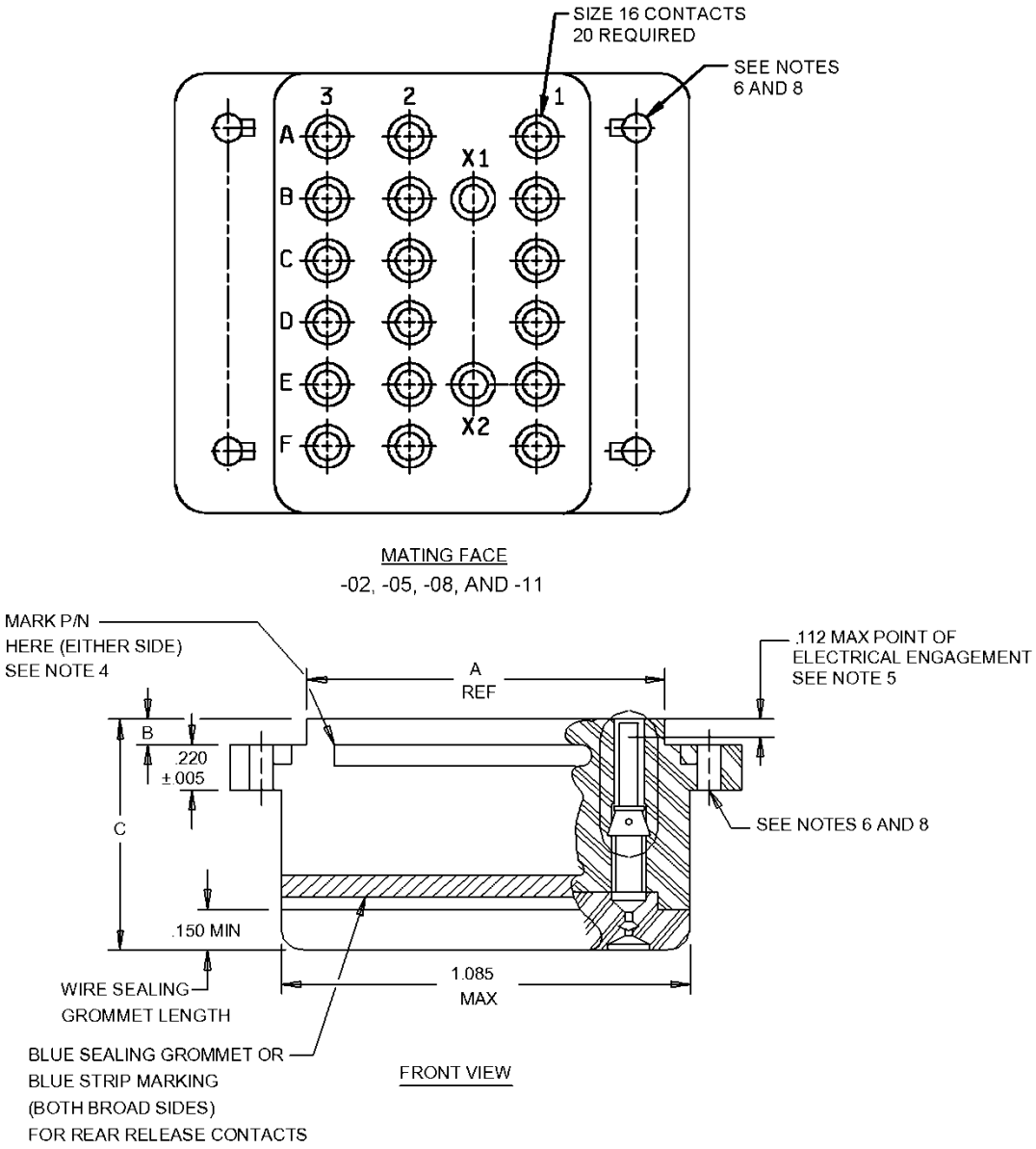

MATING FACE
-01, -04, -07, AND -10

FIGURE 1. Socket configurations.

AMSC N/A

FSC 5935

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| Dash number | A (mm) | B (mm) | C max (mm) |
|---|---|---|---|
| -01, -02, -04, and -05 | 1.075 (27.31) | .093 (2.36) | .890 (22.61) |
| -07, -08, -10, and -11 | 1.025 (26.04) | .135 - .140 (3.43 - 3.56) | 1.250 (31.75) |

FIGURE 1. Socket configurations - Continued.



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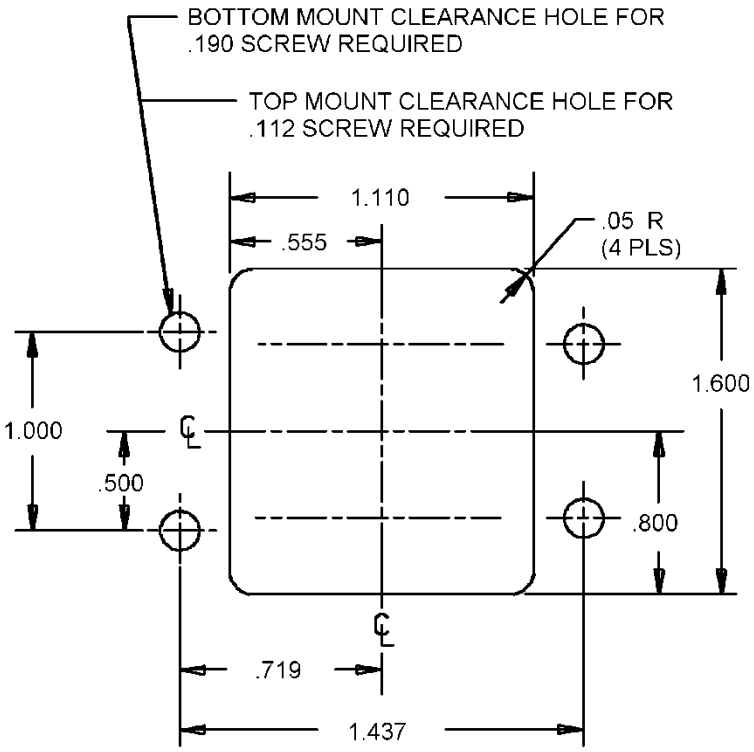
| Inches | mm | Inches | mm |
|---|---|---|---|
| .005 | 0.13 | .220 | 5.59 |
| .03 | 0.76 | .418 | 10.62 |
| .06 | 1.52 | .500 | 12.70 |
| .09 | 2.29 | 1.000 | 25.40 |
| .100 | 2.54 | 1.085 | 27.56 |
| .112 | 2.84 | 1.437 | 36.50 |
| .150 | 3.81 | 1.500 | 38.10 |
| .200 | 5.08 | 1.74 | 44.20 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are for general information only.
3. Unless otherwise specified, tolerances are ± .01 inch (0.25 mm) for two place decimals and ± .005 inch (0.13 mm) for three place decimals.
4. Marking shall be characters, which are molded .035 inch (0.89 mm) minimum. Ink marking optional in accordance with MIL-STD-1285.
5. Point of electrical contact from mating face of socket insulator to the socket contact.
6. Keyway is shown for loose stud mounting configuration only (see figure 4, detail A).
7. For mating relay (see table I).
8. Configuration for mounting, see figures 3, 4, and table I.

FIGURE 1. Socket configurations - Continued.



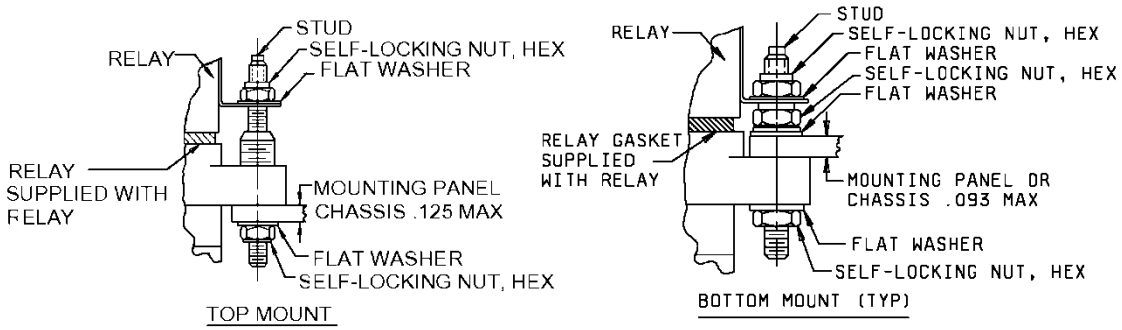
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| Inches | mm |
|---|---|
| .05 | 1.27 |
| .112 | 2.84 |
| .190 | 4.83 |
| .500 | 12.70 |
| .555 | 14.10 |
| .719 | 18.26 |
| .800 | 20.32 |
| 1.000 | 25.40 |
| 1.110 | 28.19 |
| 1.437 | 36.50 |
| 1.600 | 40.64 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are for general information only.
3. Unless otherwise specified, tolerances are ± .005 inch (0.13 mm) for three place decimals and ± .01 inch (0.25 mm) for two place decimals.

FIGURE 2. Recommended mounting dimensions.

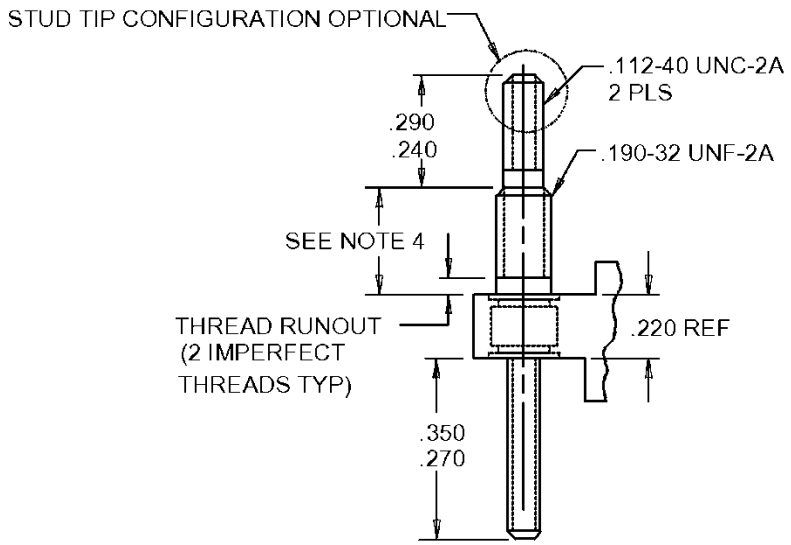



| Inches | mm |
|---|---|
| .093 | 2.36 |
| .125 | 3.18 |

FIGURE 3. Socket mounting, fixed and loose studs.



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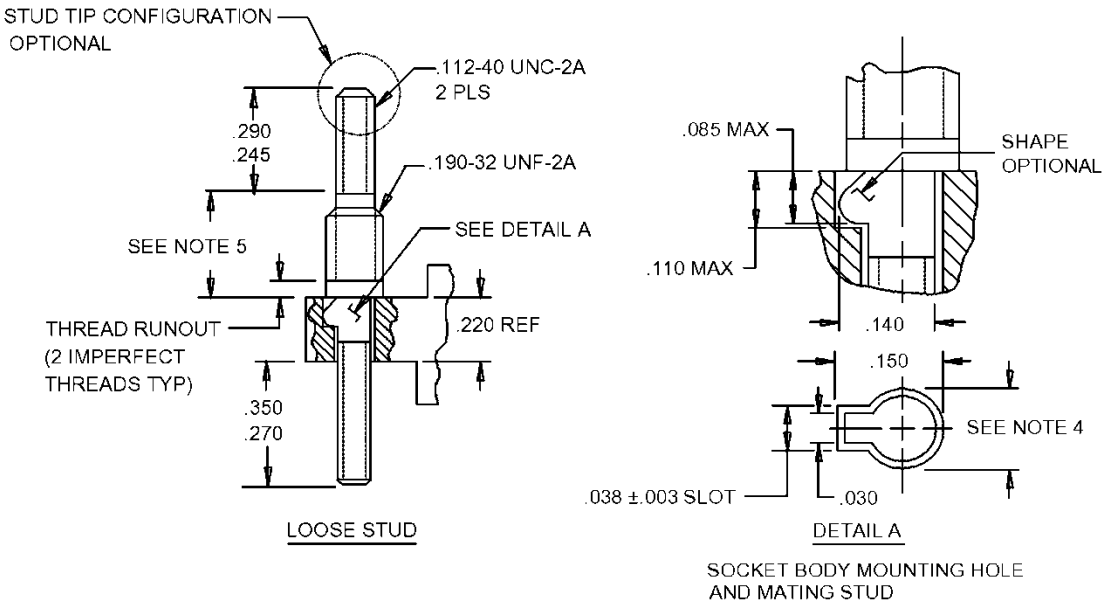

| Inches | mm |
|---|---|
| .05 | 1.27 |
| .093 | 2.36 |
| .112 | 2.84 |
| .125 | 3.18 |
| .190 | 4.83 |
| .220 | 5.59 |
| .240 | 6.10 |
| .270 | 6.86 |
| .290 | 7.37 |
| .350 | 8.89 |
| .500 | 12.70 |
| .555 | 14.10 |
| .719 | 18.26 |
| .800 | 20.32 |
| 1.000 | 25.40 |
| 1.110 | 28.19 |
| 1.437 | 36.50 |
| 1.600 | 40.64 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are for general information only.
3. Unless otherwise specified, tolerances are ± .005 inch (0.13 mm) for three place decimals and ± .01 inch (0.25 mm) for two place decimals.
4. Length from top of socket flange to base of .112-40 UNC-2A thread:
   -10 and –11 length shall be .318 ± .010 inch (8.08 ± .0.25 mm).
   -04 and –05 length shall be .290 ± .010 inch (7.37 ± 0.25 mm).

FIGURE 4. Studs fixed.



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| Inches | mm | Inches | mm |
|---|---|---|---|
| .003 | 0.08 | .150 | 3.81 |
| .030 | 0.76 | .190 | 4.83 |
| .038 | 0.97 | .220 | 5.59 |
| .085 | 2.16 | .245 | 6.22 |
| .110 | 2.79 | .270 | 6.86 |
| .112 | 2.84 | .290 | 7.37 |
| .140 | 3.56 | .350 | 8.89 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are for general information only.
3. Unless otherwise specified, tolerances are ± .005 inch (0.13 mm) for three place decimals and ± .01 inch (0.25 mm) for two place decimals.
4. The diameter for the loose stud mating hole in the socket body shall be:
   -01, and -02, diameter shall be .116 ± .003 inch (2.95 ± 0.08 mm).
   -07, and -08, diameter shall be .125 ± .003 inch (3.18 ± 0.08 mm).
5. Length from top of socket flange to base of .112-40 UNC-2A thread:
   -01, -02, -04, -05 length shall be .290 inch (7.37 mm.).
   -07, -08, -10, -11 length shall be .318 inch (8.08 mm).

FIGURE 5. Studs loose.

REQUIREMENTS:

Design and construction: See figures 1, 2, 4, 5, and table I.

Insulator: Diallyl phthalate, in accordance with ASTM-D5948, type SDG-F, or any glass filled thermoplastic material in accordance with ASTM-D5204.

Color: Material color shall be optional providing that the color provides a contrasting background for



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the blue sealing grommet or blue color bands indicating rear release contacts.

Grommet: Silicone rubber.

Mounting hardware: Corrosion resistant steel or steel with cadmium/chromate finish.

Electrical:

Insulation resistance: 1000 megohms minimum. Test pin diameter, size 16, .0625 ± .0010 inch (1.588 ± 0.025 mm).

Dielectric withstanding voltage:

Sea level: Test voltage, 1500 V rms. Test pin diameter, size 16, .0625 ± .0010 inch (1.588 ±0.025 mm).

High altitude 80,000 feet (24.4 km): Test voltage, 500 V rms. Test pin diameter, size 16, .0625 ± .0010 inch (1.588 ± 0.025 mm). For purpose of this test an air pressure of 26 millibar (2.6 kilopascals), will be used to simulate an altitude of 80,000 feet (24.4 km).

Contacts: Contacts shall be removable crimp type in accordance with SAE-AS39029/92, SAE-AS39029/5 (see table I).

TABLE I. Dash numbers and configurations.

| Dash number | Mounting style | Contact size | | Number of contacts | Contact designation M39029 | Mating relay M83536 | Current type |
|---|---|---|---|---|---|---|---|
| | | Mating end | Wire barrel | | | | |
| 01 | Loose stud (figure 4) | 16 | 16 | 20 | /92-533 | /25-002, /26-002 | DC |
| 02 | Loose stud (figure 4) | 16 | 16 | 20 | /92-533 | /27-002 | AC |
| 04 | Fixed stud (figure 3) | 16 | 16 | 20 | /92-533 | /25-002, /26-002 | DC |
| 05 | Fixed stud (figure 3) | 16 | 16 | 20 | /92-533 | /27-002 | AC |
| 07 | Loose stud (figure 4) | 16 | 16 | 20 | /5-116 | /25-002, /26-002 | DC |
| 08 | Loose stud (figure 4) | 16 | 16 | 20 | /5-116 | /27-002 | AC |
| 10 | Fixed stud (figure 3) | 16 | 16 | 20 | /5-116 | /25-002, /26-002 | DC |
| 11 | Fixed stud (figure 3) | 16 | 16 | 20 | /5-116 | /27-002 | AC |

Environmental:

Temperature range: Operating temperature range: -70 degrees C to +125 degrees C.

Wire sealing: A resilient grommet is permanently bonded to the wire entry face of the socket as to



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provide moisture sealing capabilities of from .065 inch (1.65 mm) min diameter to .109 inch (2.27mm) max diameter insulated wire.

Mechanical:

Vibration (sinusoidal): In accordance with MIL-STD-202-204, test condition G.

a. Except that the frequency range shall be varied logarithmically between the limits of 10 Hz and 3,000 Hz.

b. Except that the procedure of MIL-STD-202-201 may be applied during 10 Hz to 55 Hz band of the vibration frequency range.

c. Qualified mating relay shall be used as a test gauge.

Vibration (random): In accordance with EIA-364-28, test condition V, letter G, with a test duration 15 minutes. The mating relay shall be used as the test gauge.

Shock (mechanical): In accordance with MIL-STD-202-213, condition C, except peak values shall be 200g's.

Insertion and withdrawal forces: The insertion and withdrawal forces of the relay and socket shall be as specified in table II.

TABLE II. Insertion and withdrawal force.

| Condition | Inspection | Insertion force |
|---|---|---|
| Initial | Insertion force (max) | 30 lbf (133 newton) |
| | Withdrawal force (min) | 2.0 lbf (8.9 newton) |
| After 10 insertions and withdrawals; before vibration | Insertion force (max) | 33 lbf (147 newton) |
| | Withdrawal force (min) | 1.9 lbf (8.5 newton) |
| After vibration | Insertion force | 33 lbf (147 newton |
| | Withdrawal force (min) | 1.9 lbf (8.5 newton) |

Mounting hardware: The mounting hardware shall allow mounting the socket above, or below the panel or chassis, and shall allow mounting and securing the relay to the socket without distributing the mounted socket or access to the wiring side of the socket (see figures 3 and 4). The hardware shall provide the nominal spacing between socket surface and relay mounting flange, regardless of the mounting configuration.

Supplied with relay socket:

8 each .112-40 self locking nuts (.206 max dia x .176 max height).

8 each .112 flat washers (.224 max O.D. x .021 max thick).

4 each .190-32 self locking nuts (.330 max dia. X .190 max height).

4 each .190 flat washers (.360 max O.D. X .036 max thick).

4 each studs for loose mounting configuration only (see figure 4).



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Fixed mounting studs: Studs shall be fixed into the mounting flange of the socket and shall be designed so as to prevent rotation of the stud within the flange (see figure 3).

Contact installation tools: See table III.

TABLE III. Contact installation tools.

| Nomenclature | Part or Identifying Number (PIN) | |
|---|---|---|
| Contact type | M39029/5-116 | M39029/92-533 |
| Crimp tool | M22520/1-01, /7-01 | M22520/1-01, /7-01 |
| Positioner | M22520/1-02, /7-03 | M22520/1-02, /7-03 |
| Insertion tool | M81969/8-07 | M81969/8-07, /14-03 |
| Removal tool<br>Wired<br>Unwired | <br>M81969/14-03<br>M81969/30-06, /30-05 | <br>M81969/8-08, /14-03<br>M81969/30-06, /30-05 |

Torque: Relay socket and hardware shall be subjected to torque testing as specified in table IV. Sockets shall be installed in mounting panel when test torque is applied. No visual evidence of physical damage shall be permitted. Torque shall be maintained for a reasonable period of time to insure stud, socket, and associated hardware have not been damaged (see table IV).

TABLE IV. Torque requirements (installed in panel conditions).

| Thread size | Torque | | | |
|---|---|---|---|---|
| | Testing | | Installation | |
| | Inch-pounds | Newton | Inch-pounds | Newton |
| .112-40 | 8 +1 -0 | 0.90 + 0.11 -0 | 4 ±1 | 0.45 ± 0.11 |
| .190-32 | 24 +1 -0 | 2.71 + 0.11 -0 | 18 ±1 | 2.03 ± 0.11 |

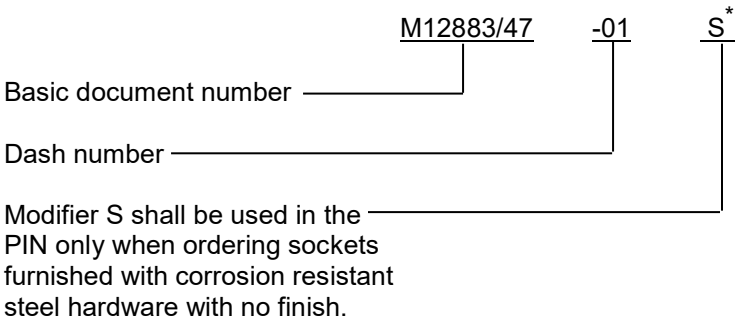
Weight: .176 pound (79.8 grams) maximum weight of relay socket, all contacts and associated hardware.

PIN: The PIN shall be marked on the socket body as shown in the example (see figure 1). The PIN shall consist of the basic number of this specification sheet, the dash number from table I, and an optional modifier.



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Example:




* For future acquisition of these sockets as of the effective date of revision D, 20 December 1989, parts with an "S" modifier shall be corrosion resisting steel (CRS), and parts without an "S" modifier shall be cadmium chromate finish. No mixing.

Ordering data: Sockets without contacts may be ordered when so indicated (see MIL-DTL-12883). This applies only to the original equipment manufacturers (OEM's) and subcontractors. All direct shipments to the government shall include all applicable contacts and mounting hardware. The PIN to be marked on the socket shall be as shown in the PIN example.

The Government PIN, specified in table V, supersedes the following commercial PINs.

TABLE V. Supersession and cross reference.

| Active Government PIN | Superseded PIN |
|---|---|
| | CAGE 58982 |
| M12883/47-01 | RSE116755 |
| M12883/47-02 | RSE116757 |
| M12883/47-04 | RSE116761 |
| M12883/47-05 | RSE116763 |
| M12883/47-07 | RSE116101 |
| M12883/47-08 | RSE116103 |
| M12883/47-10 | RSE116073 |
| M12883/47-11 | RSE116075 |

Amendment notations: The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.



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Referenced documents: In addition to MIL-DTL-12883, this document references the following:

MIL-STD-202-201
MIL-STD-202-204
MIL-STD-202-213
MIL-STD-1285
MIL-PRF-6106
MIL-PRF-83536
ASTM-D5204
ASTM-D5948
EIA-364-28
SAE-AS39029/5
SAE-AS39029/92

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5935-2020-020)

Review activities:
Army - AR, AT
Navy - AS, MC, OS, SH

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